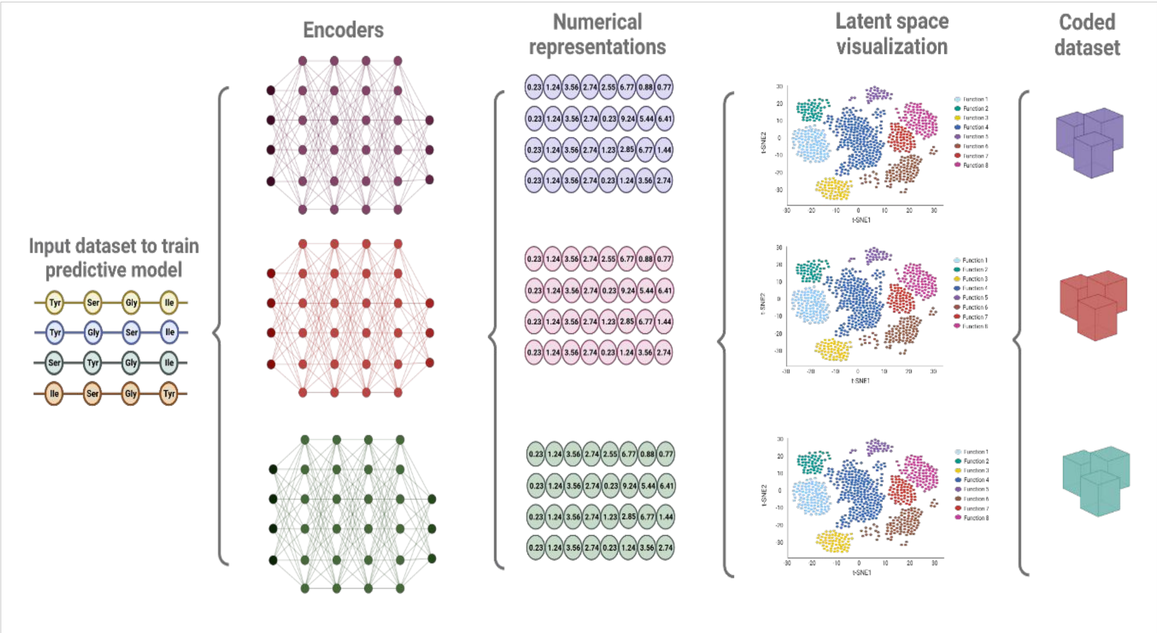
Developing sequence-based predictive models for protein-ligand and protein-protein interactions using machine learning strategies

David Medina Ortiz[1,2], Pedro Salinas[3], Fabio Durán-Verdugo[3], Álvaro Olivera-Nappa[2], Roberto Uribe-Paredes[1]
[1]Departamento de Ingeniería en Computación, Universidad de Magallanes, [2]Centre for Biotechnology and Bioengineering, Universidad de Chile, [3]Escuela de Ingeniería en Bioinformática, Universidad de Talca
Contact: david.medina@umag.cl or david.medina@cebib.cl

Motivation and objectives

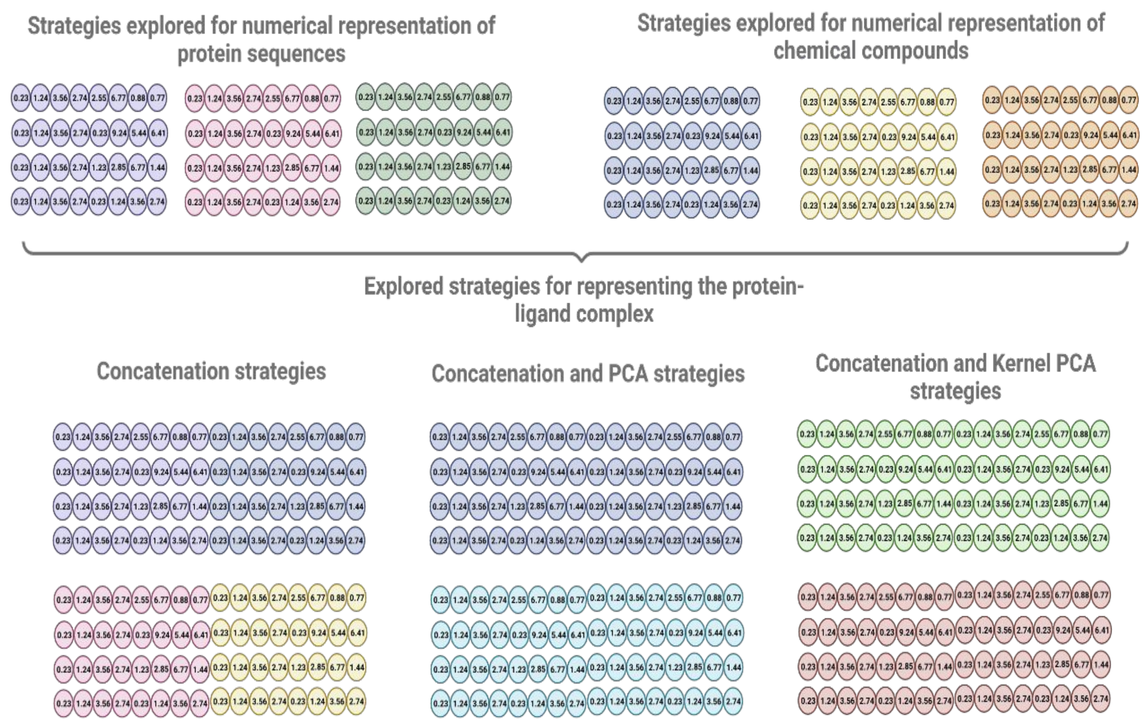
Predicting the affinity between two proteins is one of the most relevant challenges in bioinformatics and one of the most useful for biotechnological and pharmaceutical applications. Current prediction methods use the structural information of the interaction complexes. However, predicting the structure of proteins requires enormous computational costs. Machine learning methods emerge as an alternative to this bioinformatics challenge. Nevertheless, exploring several alternatives for processing and developing predictive model training is necessary.

Numerical representation strategies of proteins and chemical compounds



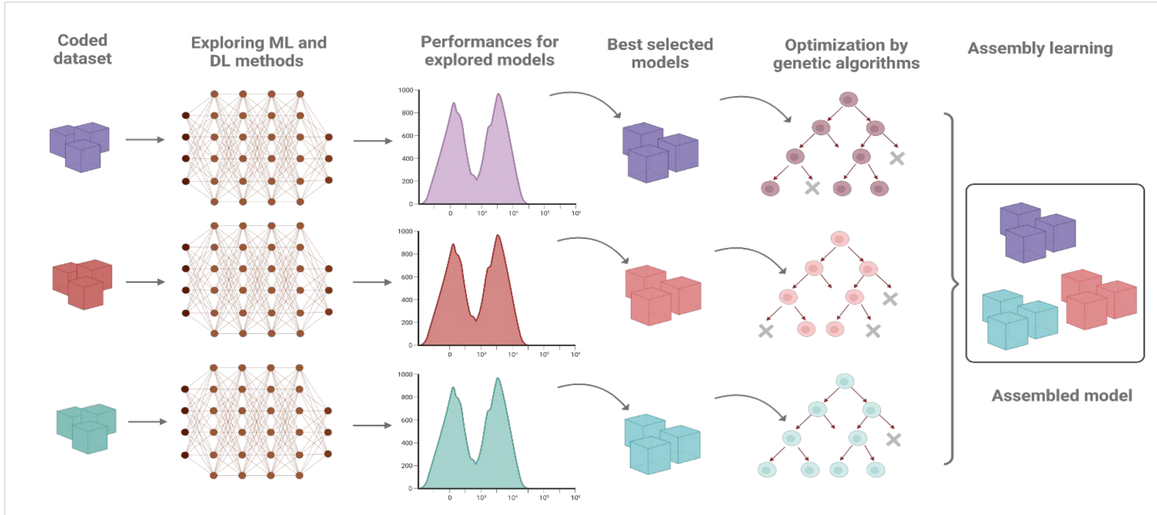
Different numerical representation strategies were explored for both protein sequences and chemical compounds. In the case of protein sequences, amino acid coding methods were explored via physicochemical properties, signal space transformations via FFT, and application of large language models via pre-trained models. While to represent the chemical compounds, methods based on characterizations via RDKit or Mordred and pre-trained models such as Mol2Vec were used.

Protein-ligand complex representation



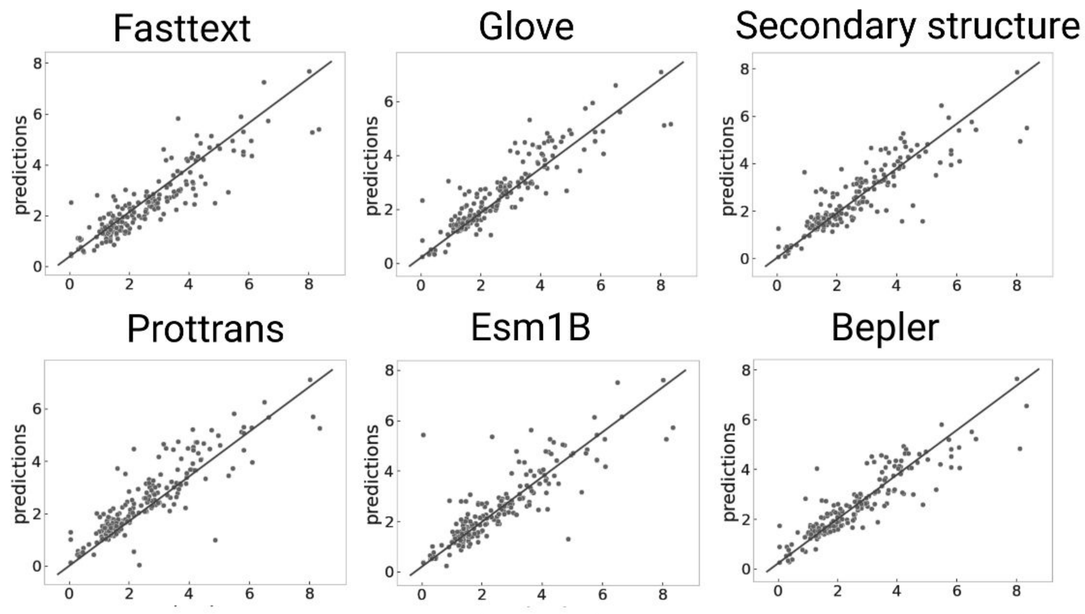
Once the proteins and chemical compounds are encoded linearly, strategies of concatenation and emulation of the structural complexes via representations of PCA or Kernel PCA are applied. In this way, three representations are created for each protein and ligand representation combination to generate data sets and develop the predictive models.

Training strategies



The training of predictive models was based on applying machine learning methods and deep learning architectures to explore predictive modeling strategies, evaluated via classic performance metrics such as the Pearson coefficient. Then, statistical methods are used to select the best combinations of algorithms and numerical representations, whose hyperparameters are optimized in a tuning process via genetic algorithms. Finally, ensemble learning strategies combine the hyperparameter-optimized models into a single predictive system.

Exploring sequence-based approaches to protein ligand affinity prediction



The best strategies to represent interaction complexes were combining large language models for proteins and mol2vec for chemical compounds combined via kernel-PCA methods. The performances of the optimized models reach a 0.7 Pearson coefficient. However, the assembled model achieves a 0.82 Pearson coefficient.

Exploring sequence-based approaches to protein networks predictive systems

Task	Training strategy	Description	R2 performance
Proximiate-Dg	ML classic	OneHot as pretrained model and Random Forest as supervised learning algorithm	0.92
	CNN methods	**Hydropathy as encoder combined with PCA representations and using A architecture**	**0.94**
	GCN methods	Alpha structure combined with FFT transformation and 15 hidden layers in the created architecture	0.31
PROXiMATE-Kon	**ML classic**	**Bepler as pretrained model and Decision tree as supervised learning algorithm**	**0.72**
	CNN methods	Alpha structure as encoder combined with FFT representations and kernel-PCA as transformation and using D architecture	0.41
	GCN methods	Secondary structure as encoder and 15 hidden layers in the created architecture	0.01
SKEMPI-Koff	**ML classic**	**Alpha structure property as encoder combined with PCA transformation and Random Forest as supervised learning**	**0.85**
	CNN methods	Volume as encoder combined with FFT representations and PCA transformations and using D architecture	0.76
	GCN methods	Beta structure as encoder and 10 hidden layers in the created architecture	0.76
PROXiMATE-Kd	ML classic	Glove as pretrained model and Decision tree as supervised learning algorithm	0.76
	CNN methods	Bepler as pretrained models and architecture A to train the predictive model	0.26
	GCN methods	Bepler as pretrained model and 20 hidden layers in the created architecture	**0.88**
SKEMPI-Affinity	ML classic	Encoder Hydrophobicity combined with FFT applications and AdaBoost as supervised learning algorithm	0.05
	CNN methods	Fasttext as pretrained model and architecture A to train the predictive model	0.01
	GCN methods	**Secondary structure as encoder and 20 hidden layers in the created architecture**	**0.87**
SKEMPI-Kon	ML classic	Bepler pretrained models and KNN as supervised learning algorithm,	0.64
	CNN methods	**Secondary structure as encoder combined with FFT applications and kernel-PCA transformation and using the architecture C**	**0.72**
	GCN methods	Energetic as encoder strategy and 30 hidden layers in the created architecture	0.01

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